# PRINCE LUNGA MTHETHWA

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## PROFESSIONAL SUMMARY

Full Stack Developer with 3+ years of experience building scalable web applications and embedded systems. Proficient in Python, Django, JavaScript, React, Node.js, Java, and C/C++. Expertise in RESTful API development, microservices architecture, database design (MySQL, MongoDB), and cloud computing. CCNA certified with strong background in IT infrastructure, network security, and industrial automation (PLC, SCADA, Arduino, ESP32).

## EDUCATION

**Diploma in Computer Systems Engineering (Software Engineering and Information Technology)**

*Vaal University of Technology, Department of Computer Science | 2020 – 2024*

- Capstone Project: Embedded Digital Sign-in System with Facial Recognition (Grade: Excellent)

**AI Software Developer (NQF Level 5) – Awaiting Graduation**

*Initium Venture*

## CERTIFICATIONS

- Microsoft 365 Fundamentals (MS-900)
- Microsoft Endpoint Administrator (MD-102)
- Cisco Certified Network Associate (CCNA) v7: Enterprise Networking, Security, and Automation
- Cisco Certified Network Associate (CCNA) v7: Introduction to Networks

## TECHNICAL SKILLS

**Programming Languages:** Python, Java, JavaScript, C, C++, Embedded C, HTML5, CSS3, SQL

**Web Development:** Full-Stack Development, Frontend Development, Backend Development, Django, Flask, Node.js, Express.js, React, Angular, Bootstrap, Responsive Web Design, RESTful APIs, GraphQL, Microservices Architecture, MVC Architecture, JSON, AJAX, XML, Web Application Security, Authentication, Authorization

**Databases:** MySQL, MongoDB, SQL Server, Database Design, Query Optimization

**Embedded Systems:** Arduino, ESP32, PIC Microcontroller, PLC (Siemens, Allen Bradley), SCADA, Circuit Design, PCB Layout, IoT, Industrial Automation

**IT Infrastructure:** Microsoft 365, Windows Server, Linux Server, Network Security, TCP/IP, DNS, DHCP, VPN, Active Directory, Virtualization

**Tools & Methodologies:** Git, GitHub, Agile, Scrum, OOP, Data Structures, Algorithms, Design Patterns, CI/CD, Docker, SDLC

## PROFESSIONAL EXPERIENCE

### Full Stack Developer

*TNG Solutions | Midrand, South Africa | November 2025 – Present*

- Develop and maintain 5+ full-stack applications using Python/Django, JavaScript/React, and Node.js with 99.9% uptime
- Implement RESTful APIs and microservices, reducing API response time by 40% through optimization

### Electrical and IT Systems Engineering Intern

*Resolution Circle | Johannesburg, South Africa | July 2024 – August 2025*

- Maintained enterprise IT infrastructure for 200+ users across 15+ facilities, achieving 99% uptime
- Developed 10+ embedded automation solutions using Arduino/ESP32 and PLC programming, reducing manual intervention by 75%

### Computer Technician / IT Support Specialist

*Trinity Solutions | Vanderbijlpark, South Africa | January 2023 – December 2023*

- Provided IT support to 100+ users across 5 departments with 95% first-call resolution rate
- Installed and configured 50+ workstations (Windows, macOS, Linux) 30% faster than estimated timelines

## TECHNICAL PROJECTS

**AI-Powered Attendance Management System |** *Full-Stack Web Application*

- Built comprehensive attendance system using React 18, TypeScript, Node.js, and MySQL with TensorFlow.js facial recognition and conversational AI chatbot (DeepSeek v3.1) for natural language queries
- Implemented real-time tracking with location verification, role-based access control, leave management, and automated reporting with interactive analytics

**Kago Human Capital (KagoHC) |** *Full-Stack HR Management System*

- Developed comprehensive HR platform with dual portals (admin/employee) covering full employee lifecycle including SARS-compliant payroll, performance management (KPIs/OKRs), leave/attendance tracking, and asset management
- Built using PHP backend with MySQL/MariaDB, deployed to production via cPanel automation with real-time analytics dashboard

## REFERENCES

*Available upon request*